Abiram Srikanthasegaram

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PROFESSIONAL SUMMARY

Computer Systems Engineering undergraduate at SLIIT (2023–2027) specialising in bridging hardware and software — from register-level AVR Assembly firmware and closed-loop PID motor control, to custom PCB fabrication and fixed-wing aircraft prototyping. Proven ability to take systems from concept to tested hardware: six complete projects delivered end-to-end, each validated against measurable performance targets. Seeking an internship in Embedded Systems, Firmware Development, Control Systems or IoT Engineering.

KEY SKILLS

Embedded Firmware AVR Assembly · C/C++ · Arduino · ATmega328P · Register-level I/O, PWM, ADC, Timer configuration · I2C · SPI · UART · STM32 (in progress)

Control Systems PID / PI control · MATLAB · Simulink · Closed-loop design · Encoder feedback · Motor control · FreeRTOS (familiar)

Hardware Design PCB Design (KiCad, EasyEDA) · 3D CAD & Enclosure Design (Fusion 360) · Two-layer board fabrication · Soldering · DRC · Signal integrity

Platforms & SoC Raspberry Pi 4 · ESP32 · Arduino Uno · L298N motor driver · IR sensors & receivers

Software & Tools Python · Git · Linux · GPIO programming · Real-time GUI (Tkinter) · CSV data logging

Aeronautics Fixed-wing airframe design · Airfoil & CG analysis · Thrust-to-weight optimisation · XFLR5 · OpenVSP · NASA Glenn simulations

Electronics Digital logic · Analog circuits · BJT biasing · RC filters · Oscilloscope · Decoupling & noise filtering

ENGINEERING PROJECT EXPERIENCE

Embedded Systems Project Engineer — *Nov 2023 – Present*

SLIIT – Engineering Projects | Colombo, Sri Lanka

- Designed and tuned a closed-loop PI controller (Kp=1, Ki=0.45) on Raspberry Pi — bringing a 12V DC motor from rest to setpoint with overshoot below 15% and steady-state error under 2%, validated in MATLAB/Simulink before committing to hardware, with a custom two-layer PCB fabricated to replace the breadboard prototype.
- Programmed a fully autonomous mobile robot entirely in AVR Assembly with zero library dependencies — all peripheral configuration (DDR/PORT/PIN registers, Timer0 Fast PWM, ADC) written at register level, implementing an 8-state FSM achieving 100% pass rate across all 5 line-following and parking test scenarios.
- Designed and built a complete end-to-end IoT security pipeline (CLIMS) on ESP32 — PIR motion trigger → camera capture → Python facial recognition → electronic lock actuation → real-time mobile push notification, engineered to fire alerts only on confirmed unrecognised faces, not general motion.
- Completed the full PCB lifecycle for a 1-digit asynchronous up counter — EasyEDA schematic capture → DRC sign-off → board fabrication → soldering → oscilloscope validation of switching waveforms at each 74HC74 flip-flop stage.
- Built a configurable trace-driven cache simulator in Python sweeping 36 parameter combinations, identifying 64B as the optimal cache line size with a 35% miss ratio reduction vs 16B, and AMAT of ~13.0 ns at the optimal L1 configuration.

ACADEMIC PROJECTS

PID-Controlled DC Motor Stabilisation System — *Year 3, Sem 1 – 2026*

Raspberry Pi 4 · Python · L298N H-Bridge · Optical Encoder · MATLAB/Simulink · EasyEDA · Two-Layer PCB

- Designed and tuned a closed-loop PI controller (Kp=1, Ki=0.45) that brought a 12V DC motor from rest to setpoint with overshoot below 15% and steady-state error under 2%, fully satisfying the design specification — validated in MATLAB/Simulink simulation before committing to hardware.
- Fabricated a custom two-layer PCB to replace breadboard wiring, isolating encoder signal paths from high-current motor traces — directly eliminated the intermittent encoder noise and voltage-drop instability that caused controller divergence during prototype testing.
- Built a real-time Python monitoring application featuring live RPM graphing, dynamic Kp/Ki adjustment sliders, PWM duty cycle display, and automated CSV logging — enabling iterative tuning without reflashing firmware.
- Made a deliberate engineering decision to omit the derivative term: experimental testing confirmed Kd amplified encoder quantisation noise, introducing control output jitter without improving rise time or reducing overshoot — a PI-only design was justified and documented.
- Implemented interrupt-driven encoder pulse counting on Raspberry Pi GPIO with separate power rails for logic and motor circuits, preventing motor switching transients from corrupting speed measurements.

Line-Following Arduino Car with Autonomous Parking — *Year 3, Sem 1 – 2026*

AVR Assembly · ATmega328P · Arduino Uno · L298N · 3× IR Line Sensors · 38 kHz IR Receiver

- Programmed a fully autonomous mobile robot entirely in AVR Assembly with zero library dependencies — all peripheral configuration (DDR/PORT/PIN registers, Timer0 Fast PWM, ADC) written at register level, demonstrating deep understanding of the ATmega328P architecture.
- Designed an 8-state finite state machine (*Initialization → Line Following → Marker Detection → Slow Mode → Parking Detection → Parking Execution → Lost Line Recovery → Stop*) that handled every operational scenario tested, achieving a 100% pass rate across all 5 test scenarios.

- Engineered analog IR parking detection using ADC threshold comparison on channel A0, triggering a deterministic 7-step parking manoeuvre sequence — chosen over digital detection to provide a configurable sensitivity margin against ambient IR interference.
- Implemented a lost-line recovery routine performing alternating left/right swing searches with bounded retry counts — preventing the robot from driving indefinitely off-track during sharp turns or surface glare events.

Cache Design & Performance Analysis Simulator *Year 2 – 2025*

Python · Computer Architecture · Trace-Driven Simulation · AMAT Analysis

- Built a configurable trace-driven cache simulator from scratch in Python, sweeping 36 parameter combinations across line sizes (16B, 32B, 64B, 128B), associativity (1-way, 2-way, 4-way), and replacement policies (LRU, FIFO, Random) — grounded in six testable hypotheses derived from IEEE/ACM literature review.
- Identified 64B as the optimal cache line size: simulation showed miss ratio improving from 0.237 at 16B to 0.155 at 64B — a 35% reduction — before degrading at 128B due to over-fetch and cache pollution, directly confirming the spatial locality sweet-spot hypothesis.
- Quantified that LRU, FIFO, and Random replacement policies converged to nearly identical miss ratios (~0.1806) at steady state — concluding that policy selection is workload-dependent and over-engineering replacement logic is unwarranted for workloads with low capacity pressure.
- Determined the optimal L1 configuration: 64B lines, 2-way associativity, LRU policy achieving an AMAT of ~13.0 ns — aligning with industry-standard L1 design practice.

CLIMS – Smart Surveillance & Access Control System *Year 1 – 2024*

ESP32 · IoT · Python · OpenCV Facial Recognition · Electronic Door Lock · Mobile Push Notifications

- Designed and built a complete end-to-end IoT security pipeline on ESP32: PIR motion trigger → camera capture → Python facial recognition → electronic lock actuation → real-time mobile push notification, with the full decision workflow executing on a single embedded platform.
- Engineered the system to distinguish between known and unknown faces before triggering the lock — the only action that fires an alert is a confirmed unrecognised face, not general movement, preventing false positives from ambient motion.

1-Digit Asynchronous Up Counter – PCB Design & Fabrication *Year 2 – 2025*

EasyEDA · 74HC74 D Flip-Flops · Seven-Segment Display · PCB Fabrication · Oscilloscope Validation

- Completed the full hardware lifecycle for a 0–9 asynchronous decimal counter — EasyEDA schematic capture → multi-layer PCB layout → DRC sign-off → board fabrication → component soldering → oscilloscope validation of switching waveforms at each flip-flop stage.
- Verified correct decade counting behaviour and clean logic transitions on oscilloscope, confirming that the physical board matched the simulated timing behaviour.

PERSONAL PROJECTS

Agni Wing – Fixed-Wing Aircraft Prototype *2025 – Ongoing*

Airframe Design · A2212 1125KV BLDC · 40A ESC · 3S 2200mAh LiPo · 1045/1047 Propeller · Fusion 360 CAD · NASA Glenn Simulations

- Co-founded a four-person self-directed engineering team (Agni Wing) with the long-term goal of building an autonomous fixed-wing UAV — beginning with a manually controlled prototype to validate aerodynamic fundamentals before integrating flight control electronics.
- Completed the full airframe design cycle within 6 days: aerodynamic layout, fuselage sizing, CG calculation, propeller offset, thrust-line alignment, and wing-fuselage integration — using NASA Glenn's Beginner's Guide to Aeronautics for airfoil and chord analysis, and Fusion 360 CAD for dimensional definition.
- Ran 5 iterative build-test-crash-analyse-rebuild cycles — the aircraft achieved approximately 12 seconds of sustained flight reaching 30–40 ft altitude on Attempt 5, validating that the CG positioning and thrust-to-weight ratio (AUW ~850–900 g) were within flyable margins.
- Each failed attempt produced a structured failure analysis: Attempts 1–3 identified CG instability and insufficient thrust margin; Attempt 4 revealed structural weakness at the wing root under load; Attempt 5 incorporated all corrections and achieved the first controlled flight.

CERTIFICATIONS & TRAINING

PLC & Industrial Automation Workshop (PLCFI) *2026*

IEEE Student Branch, Curtin University Colombo · Sri Lanka Institute of Robotics (SLIR)

- Completed hands-on sessions covering PLC fundamentals, ladder logic programming, HMI/SCADA configuration using XINJE PLC hardware, and live motor and sensor control demonstrations. Awarded certificate of completion by SLIR.

EDUCATION

BSc (Hons) Computer Systems Engineering *Nov 2023 – Nov 2027 (Expected)*

Sri Lanka Institute of Information Technology (SLIIT)

- Relevant modules: Embedded Systems Design · Control System Engineering · Digital Electronics · Computer Architecture · Analog Electronics · Systems Modelling & Prototyping

REFERENCES

Three confirmed referees — full contact details available upon request.

Mr. Dinith Primal
Head of Department – Computer Systems Engineering
MSc Wireless Communications
Sri Lanka Institute of Information Technology (SLIIT)
Relationship: Head of Department — confirmed as referee

Mr. Sanjeeva Perera
Senior Academic Fellow – Faculty of Computing
BSc Eng (University of Moratuwa) · MBA (University of Colombo)
Sri Lanka Institute of Information Technology (SLIIT)
Relationship: IE3014 Professional Skills lecturer — confirmed as referee

Ms. Dinithi S. Pandithage
Lecturer & Year 2 CSNE Coordinator – Faculty of Computing
BSc (Hons) Computer Engineering (KDU) · MIEEE · MIET
Sri Lanka Institute of Information Technology (SLIIT), Malabe Campus
Relationship: IE3064 Embedded Systems Design lecturer — supervised AVR Assembly robot project, confirmed as referee